

Yingrui Hou

Statistical Modelling & Simulation | Inference, Validation & Python

Clermont-Ferrand, France | +33 (0)785396332 | yingrui.hou@outlook.com

Website: yingrui-hou.github.io/PersonalPage

LinkedIn: linkedin.com/in/yingrui-hou | GitLab: gitlab.com/hyrsophie | CERN: gitlab.cern.ch/yihou

SUMMARY

Quantitative researcher with a PhD in experimental physics and several years of postdoctoral experience developing Python/C++ workflows for probabilistic modelling, simulation calibration, latent-parameter inference, and uncertainty-aware validation on large, noisy datasets.

Experienced in likelihood-based models, joint fits, simulation-to-data alignment, covariance-aware checks, pseudoexperiments, time-dependent inference, and reproducible analysis in international scientific collaborations.

Motivated to transfer this methodological experience to simulation-based inference and mechanistic epidemiological models for real-time epidemic intelligence.

CORE SKILLS

- Statistical inference: maximum-likelihood estimation, probabilistic and mixture models, joint fitting, latent-parameter estimation
- Simulation and calibration: Geant4, simulation-to-data alignment, response modelling, bias correction, model validation
- Uncertainty and robustness: uncertainty quantification, covariance-aware checks, pseudoexperiments, stress tests
- Data science and ML: Python, NumPy, pandas, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Time-dependent analysis: time-series modelling under noise, acceptance and resolution modelling, dataset-specific corrections
- Reproducible research: C/C++, Linux, Git, CI/CD, YAML / Shell / CMake, shared codebases, large-scale workflows

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++/ROOT workflows for time-dependent inference and weak-signal extraction from large datasets with incomplete event-level ground truth.
- Implemented probabilistic and maximum-likelihood models, including joint fits, to estimate latent parameters under background contamination, finite resolution, and biased observations.
- Built simulation-to-data calibration and reweighting workflows that integrated heterogeneous simulated, control, and measured datasets before downstream inference.
- Contributed covariance-aware validation, pseudoexperiments, and uncertainty-stress checks for advanced weighted and unbinned inference methods.
- Developed CatBoost-based pipelines with feature studies, train/test diagnostics, threshold scans, and reproducible model application under dataset shift.
- Built monitoring workflows that compared predicted and observed behaviour, identified anomalies, and translated findings into calibration or software corrective actions.
- Worked autonomously in Linux-based shared codebases and international collaborations, contributing reusable methods, technical documentation, peer-reviewed publications, and scientific presentations.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed Geant4/C++ simulation and statistical-analysis workflows combining mechanistic response modelling, calibration, signal reconstruction, and validation against experimental data.

SELECTED OUTCOMES

- Validated a novel inference workflow under background, efficiency, resolution, and modelling variations using covariance-aware checks and pseudoexperiments.
- Improved reconstructed-energy resolution by about 2x through time-resolved component modelling, statistical fitting, and calibration on realistic measurement data.

SELECTED PROJECTS

Advanced Inference Workflow Validation

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

Sensor Simulation & Reconstruction

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor response.

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2